

Ricky Ding

e0134117@u.nus.edu | [GitHub](#) | [LinkedIn](#)

Visit <https://rickyding1997.github.io> for detailed descriptions, images, certificates and updates

SKILLS

- **Languages:** TypeScript/JavaScript, Python, C++, SQL, Verilog.
- **AI & Science:** PyTorch, Numba, NumPy.
- **Backend & Systems:** Node.js, Bun, Elysia, FastAPI, PostgreSQL, Redis, Docker, Caddy.
- **Frontend & Mobile:** Vue 3, React, React Native, Capacitor, VitePress.
- **Hardware:** FPGA/CPLD, PCB Design, Circuit Design.

EDUCATION

National University of Singapore (NUS)

2016 – 2020

Bachelor of Science (Honours, Highest Distinction) in Physics

- **Scholarship:** Science & Technology Undergraduate Scholarship.
- **Thesis:** Density Potential Functional Theory (DPFT) in position and momentum space.
- Developed an orbital-free DPFT code leveraging **PyTorch** for GPU acceleration.

University of Göttingen, Germany

Summer 2019

Non-Degree Study Abroad (Master of Science Physics coursework)

- **GPA:** 1.4 / 1.0 (German scale: Very Good).
- Key Coursework: Biomolecular Physics and Simulations, Hard Problems in Soft Matter, Biophysics of Gene Regulation.

EMPLOYMENT

Frontline Industrial Software Pte. Ltd

Nov 2021 – Nov 2025

CTO & Co-Founder

- Raised **USD \$700K** seed funding from Cocoon Capital and Plug and Play; participated in Entrepreneur First (2021).
- **Schedule Optimization:** Developed an original differentiable optimization engine that transforms non-differentiable project schedules into computational graphs for gradient-based optimization.
- **AI Features:** Implemented overrun forecasting using OpenAI embeddings and HDBSCAN clustering; built GPT-powered task generators.
- **Project Management:** Led technical development of the "ProjectOptimizer" AI app used in construction planning.

Centre for Quantum Technologies (CQT), NUS

July 2020 – May 2021

Researcher

- **Quantum Optimal Control:** Used Deep Reinforcement Learning (SAC) to generate control pulses for driving quantum state transitions in Transmon systems.
- **Hardware Automation:** Developed Python-based calibration procedures for IQ Mixers to minimize LO leakage and correct gain/phase imbalances via Quantum Machines hardware.

PERSONAL PROJECTS

NeuroAI Research

- **Brain Modules:** Project replicating brain module computations using ANNs to model biological intelligence through functional abstraction rather than spike-level simulation.
- **Minimalist RL:** A decoupled, lightweight reinforcement learning library (available on PyPI).
- **Gradient-Free Optimizer:** Python library for optimizing neural network parameters using genetic algorithms.

Serena Trading (Research)

- **Deep Trader:** Exploring deep learning for trading.
- **Fast Trading Simulator:** A **Numba-accelerated** minimalist simulator supporting multi-symbol positions and risk control.
- **Crypto Data Downloader/Monitor:** Tools for high-concurrency kline data retrieval and real-time market monitoring.

AI Applications & Platforms

- **EntreGames:** A conversation-based entrepreneurship game where users roleplay as CEOs/CTOs. Features AI-generated scenarios, team collaboration, and dynamic business metric tracking.
- **Similar Minds / When Light Arrives:** A social discovery platform using vector similarity search to connect users with "kindred spirits" based on deep ideological alignment. Built with Capacitor for cross-platform deployment.

Hardware & Physics Simulation

- **Musical Tesla Coil Gun:**
 - Engineered a DRSSTC system from scratch including PCB layout and hand-soldering.
 - Programmed an **EPM240 CPLD (Verilog)** for zero-current switching and audio modulation.
 - Built a **Vue3 Synthesizer App** to control the coil's energy envelope via Bluetooth using MIDI data.
- **Tesla Coil Physics Simulator:** A Python-based simulator for coupled series RLC resonator systems (available on PyPI).